



Towards an operational alignment approach for organizations

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ABSTRACT

An effective information system improves the profitability of organizations. However, it appears that there is a need for an alignment approach that allows the continued operation of information system. In our work, we focus particularly on the operational alignment. Moreover, we deal with the topic of alignment by analyzing three levels: business requirement, business process, and software system. In this paper, we present an overview of this topic's concepts. Then, we analyze approaches related to our work by defining a set of criteria. Finally, we present the main idea of our proposed approach.

CCS Concepts

• **Applied computing** → **Enterprise computing** → **Business-IT alignment**

Keywords

Business-IT Alignment; Organization; Business Requirement; Business Process; Software System; Information System; Alignment Approach.

1. INTRODUCTION

Managing information systems (IS) is an important key to success for organizations. Using information technologies to support business becomes a priority. An organization represents a set of contracts and rules that define roles and establish their relationships [1]. In the context of the organization, individuals may assume or leave roles that have been defined for general objectives or for a permanent use [1]. In this work, we define an organization as an organism in which people aim to achieve a common goal. An organization as an internal entity can provide products or services to an external entity. Therefore, it is essential for organizations to maintain a continuous operating of their components. This continuity can be ensured by aligning the business and IT domains.

Many authors dealt with alignment from various angles [2]. To propose their approach they choose particular levels to link them. In this paper, we are interested in three levels: the first level is

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business requirement which describes functions needed by any organizational unit like an employee, a department, a division or other units [3]. The second level is *business process* which refers to the definition of a collection of activities that are interdependent and which accomplish a specific business objective according to specified policies [4]. Then, the third level is *software system* which designs a set of software components and related relationships, that can implement services and functionalities and which may help to execute business activities [5]. Our purpose is to provide a method for organizations to apply the operational alignment. We choose this type of alignment based on the SAM model [6] that we develop in the related work section. Indeed, operational alignment aims to align the internal components of business and IT domains (cf. figure 1).

Moreover, we aim to analyze the operational alignment within the three levels: business requirement, business process (belonging to business domain) and software system (belonging to IT domain). Therefore, we propose a global guideline which comprises phases of alignment. Those phases are organized in this guideline according to a set of situations. Indeed, we align business requirement to business process and then business process to software system.

The rest of the paper is organized as follows: in section 2, we introduce the SAM model and approaches that deal with the topic of operational alignment. Those approaches are analyzed according to a set of criteria in section 3. In section 4, we expose our proposed approach. Finally, we conclude and present our future work.

2. RELATED WORK

2.1 The strategic alignment model

Alignment is exposed by Henderson and Venkatraman [6]. They propose the Strategic Alignment Model (SAM) which divides strategic alignment into two categories: *strategic fit* and *functional integration*.

Strategic fit is the alignment between external and internal domains. External domain comprises business strategy and IT strategy. Internal domain refers to Organizational infrastructure and processes (according to the business domain) and IS infrastructure and processes (according to the IT domain).

Functional integration concerns the alignment between business and IT domains. This category can as well be divided into two sub-categories: *strategic integration* and *operational integration*. The first sub-category aligns business strategy to IT strategy. While the second sub-category links business and IT corresponding to internal domain. Figure 1 illustrates the strategic alignment model proposed by Henderson and Venkatraman [6]. In this paper, we are interested in the operational alignment, also known as the operational integration in the SAM Model.

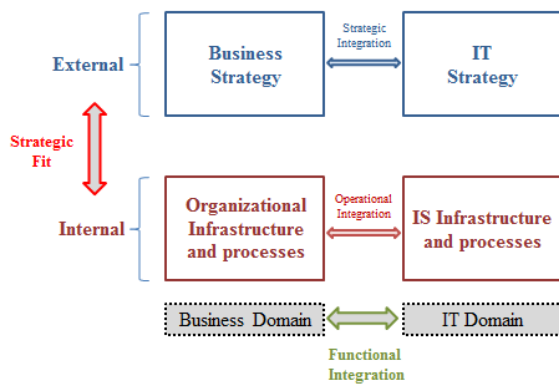


Figure 1. The SAM Model according to [6]

2.2 Operational alignment approaches

In this section, we present approaches that align at least two of the three levels: business requirement, business process, and software system.

Li et al. [7] present an approach named GSP which includes goal, scenario, and business process models. In fact, they represent goal model by the Goal Oriented Requirement Language (GRL). Scenario model is modeled by Use Case Maps (UCM). Business process model is represented using BPMN. To link the business requirement and business process levels, authors define a set of rules mapping. Those rules are represented using the QVT operational language [8]. In fact, they establish a set of rules from GRL to UCM and from UCM to BPMN.

Kraiem et al. [9] link intentional and operational levels. Indeed, they define mapping rules to translate a MAP model into a BPMN model. MAP is a directed graph in which nodes are goals. It is represented by using UML formalism. Then, authors present the mapping process to realize the transformation from MAP to BPMN.

Sousa and Julio [10] use i* to represent goal model and BPMN to model business process. Their approach of alignment consists on mapping then merging i* to BPMN. They propose the GPI (Goals, Process, and Indicators) proposal which aims to merge i*, BPMN, and KPI (Key Performance Indicators).

Frankova et al. [11] present the Business Processes with Service Level Agreements (BP&SLA) methodology. The aim is reducing the gap between early business requirements and executable business processes. For that, they carried out the following steps:

- Analyzing the organizational context using Tropos.
- Defining an hypergraph as an intermediate for reasoning about business processes and their qualities.
- Building an hierarchy of business processes using BPEL specifications.
- Building a constraint system.

Aversano et al. [5] present an approach to align business processes and software system according to three areas. The first phase is *modeling alignment* by the following steps:

- Representing business processes by using UML activity diagrams and by adding some notations.
- Representing the software system by using UML use cases, classes, states, and sequence diagrams.
- Identifying the relations existing between the two models.

- Identifying the functionalities of software that must be modified or implemented to support the business processes.

The second phase concerns *the evaluation of alignment*. In fact, the authors propose a method to codify the concept of alignment by using a set of metrics. Thus, five metrics are calculated to measure the percentage of business elements supported by the software systems and four others to specify the degree of adequacy offered by the automatic support to the business entities.

Then, the third phase is *evolution execution*. It deals with the evolution of the activities when a misalignment is detected.

Doumi et al. [12] present a metamodel for business IT alignment. They link the goal model to the business process then align business process to the information system. They use UML to model business and IT domains. i* formalism is used for representing goal model. Then, a set of metrics are proposed to evaluate the degree of alignment.

DeCastro et al. [13] exploit an approach based on MDA. They aim at specifying the correspondence between the CIM (computation independent model) and the PIM (Platform-Independent Model). Therefore, they use e3 value to represent relations and values between actors of business processes and BPMN to represent the business view. Software view is represented by UML. Indeed, they propose the following steps:

- Defining mappings between models by natural language.
- Collecting the mappings by a set of rules.
- Implementing rules by ATL language.

Elvesæter et al. [14] propose a method to align business and IT. Indeed, they propose 8 mapping rules that allow to transform a BPMN model to SoaML model. They implement the approach within CIMFlexMT which supports the model-to-model transformation using Atlas Transformation Language (ATL).

Cibran [15] proposes an approach to align business and IT by defining a set of mapping rules. Those rules aim to transform each element of BPMN to its correspondent in UML Activity metaclass. Then they implement their approach using ATL.

Han et al. [16] present an approach to align goals to services. They use the Business Motivation Model (BMM) to represent goal model and Service oriented architecture Modeling Language (SoaML) for service model.

Gehlert et al. [17] propose an approach to align business requirement and service. They use the Tropos' goal model to establish this alignment.

Wan-Kadir and Loucopoulos [18] present one aspect of the approach Manchester Business Rules Management (MBRM). In fact, they map business rules to software design elements.

3. ANALYSIS OF APPROACHES

Our objective throughout this section is to give a synthesis of approaches presented in the previous section. Therefore, we define a set of criteria. In the following, "BR" means Business Requirement, "BP" means Business Process, "SS" means Software System:

- **AL BR - BP**: this criterion identifies approaches that align business requirement and business process.
- **Aligning BP - SS**: this criterion identifies approaches that align business process and software system.
- **Aligning BR - SS**: this criterion identifies approaches that align business requirement and software system.

- **Modeling language:** this criterion specifies which languages are used to represent models of business requirement business process or software system.
- **LINK:** it specifies if the approach presents a method to link models.
- **EVAL:** it indicates if the approach presents a method to calculate the degree of alignment between levels and evaluate this alignment.
- **AMEL:** it determinates if the approach proposes a method to ameliorate models and gives a solution when a misalignment is detected.
- **TOOL:** it determinates if the work uses a technical support to implement the approach.

From this review, we note that the majority of existing approaches have taken up the issue of alignment by linking models. Two papers ([5] and [12]) present a method to evaluate alignment by calculating a set of metrics. Only one paper [5] proposes solutions to ameliorate elements that are misaligned. [12] is also the only one which is interested in linking goal models to business processes then business processes to software system.

Furthermore, we made the observation that there is no work that presents an approach taking into account the three levels (business requirement, business process, and software system) of the organizations and helping them to apply the alignment.

Table 1. Synthesis of approaches

Ref	Modeling Language			AL BR - BP	AL BP - SS	AL BR - SS	L I N K	E V A L	A M E L	T o o l
	BR	BP	SS							
[7]	GRL	BPMN	-	Y	-	-	Y	-	-	Y
[9]	MAP	BPMN	-	Y	-	-	Y	-	-	-
[10]	i*	BPMN	-	Y	-	-	Y	-	-	-
[11]	Tropos	BPEL	-	Y	-	-	Y	-	-	Y
[5]	-	UML	UML	-	Y	-	Y	Y	Y	Y
[12]	i*	UML	UML	Y	Y	-	Y	Y	-	-
[13]	-	BPMN and e3 value	UML	-	Y	-	Y	-	-	-
[14]	-	BPMN	SoaML	-	Y	-	Y	-	-	Y
[15]	-	BPMN	UML	-	Y	-	Y	-	-	Y
[16]	BMM	-	SoaML	-	-	Y	Y	-	-	-
[17]	Tropos	-	Tropos	-	-	Y	-	-	-	-
[18]	UML	-	UML	-	-	Y	Y	-	-	-

4. OUR PROPOSED IDEA

4.1 The context of patterns

The objective through our approach is to propose a guideline for organizations to help them in applying the operational alignment. For this purpose, we propose a pattern system that can serve as a guideline for the organization to achieve its alignment approach.

A process pattern was defined by Ambler [19] as a collection of techniques, actions, and/or activities for developing object-oriented software. It describes what we should do without providing the details of how we should do something [19].

In the literature, various formalisms are proposed to present patterns: the formalism of Ambler [19], the formalism P-Sigma [20], the formalism of Storell [21], RHODES [22], PROMENADE [23], the formalism of Gnat et al. [24], PROPEL [25]. Those formalisms consider the three common fields: the problem, the context, and the solution.

In this paper, we choose the P-Sigma formalism. It expresses a semantic which is common to the majority of the other formalisms [20]. In particular, we use the fields: *identification*, *classification*, *context*, *problem*, *force*, *process solution*, *requires*, and *uses*.

To apply alignment, we propose five patterns that take into consideration the phases of *mapping*, *transformation*, *verification*, *evaluation*, and *amelioration*. Then, we present the pattern *alignment* which uses all the previous patterns.

We use the following abbreviations:

- **M1:** Model 1, we use this abbreviation to express the first model we aim to align.
- **M2:** Model 2, we use this abbreviation to express the second model we aim to express.
- **MM1:** Meta-Model 1, it is the meta-model of M1.
- **MM2:** Meta-Model 2, it is the meta-model of M2.
- **TM:** Table of Matching
- **TE:** Table of Equivalent elements
- **TI:** Table of Isolated elements
- **AD:** Alignment Degree
- **MPI:** Master Plan in the Input
- **MPR:** Master Plan as Result of the alignment process
- **M1R:** Model 1 as Result of the alignment process
- **M2R:** Model 2 as Result of the alignment process
- **BRM:** Business Requirement Model.
- **BPM:** Business Process Model.
- **SSM:** Software System Model.

4.2 The proposed process patterns

In the following process patterns, M1 and M2 represent in the first case business requirement model and business process model. In the second case, they represent business process model and software system model

Table 2. Pattern of Mapping

Fields	Values
Identification	Mapping
Classification	Meta-model, matching
Context	Required M1, MM1, and MM2
Problem	We aim to map MM1 and MM2
Forces	Obtaining a table of rules to match MM1 and MM2

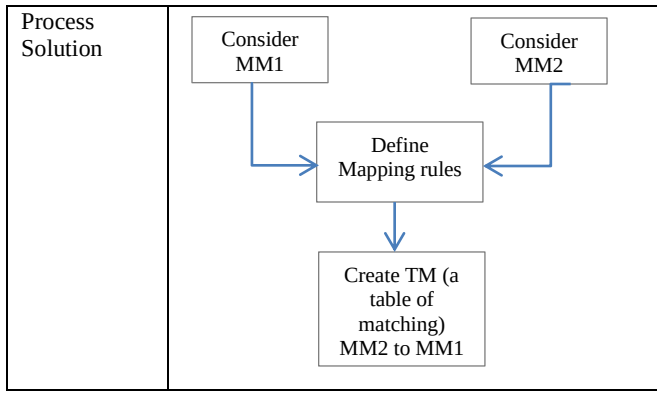


Table 3. Pattern of Transformation

Fields	Values
Identification	Transformation
Classification	Modeling
Context	Required M1 and TM, pattern of mapping
Problem	We aim to transform the model M1 to M2
Forces	Obtaining the second model
Process Solution	<pre> graph LR M1[Consider M1] -- TM --> M2[Create M2] </pre>
Requires	Pattern of Mapping

Table 4. Pattern of Verification

Fields	Values
Identification	Verification
Classification	Correspondences, equivalent elements, isolated elements
Context	Required M1, M2, pattern of mapping
Problem	We aim to identify correspondences between M1 and M2
Forces	Obtaining two tables of equivalent and isolated elements between M1 and M2
Process Solution	<pre> graph TD M1[Consider M1] --> Compare[Compare elements of M1 and M2] M2[Consider M2] --> Compare Compare --> TE[Create TE (a table of equivalent elements)] Compare --> TI[Create TI (a table of isolated elements)] </pre>

Requires	Pattern of Mapping Pattern of Transformation
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Table 5. Pattern of Evaluation

Fields	Values
Identification	Evaluation
Classification	Alignment Degree
Context	Required TE, TI, pattern of verification
Problem	We aim to evaluate alignment between M1 and M2
Forces	Calculating metrics and alignment degree
Process Solution	<pre> graph TD TE[Consider TE] --> Metrics[Calculate the metrics to evaluate alignment] TI[Consider TI] --> Metrics Metrics --> AD[Identify the degree of alignment (AD)] </pre>
Requires	Pattern of Verification

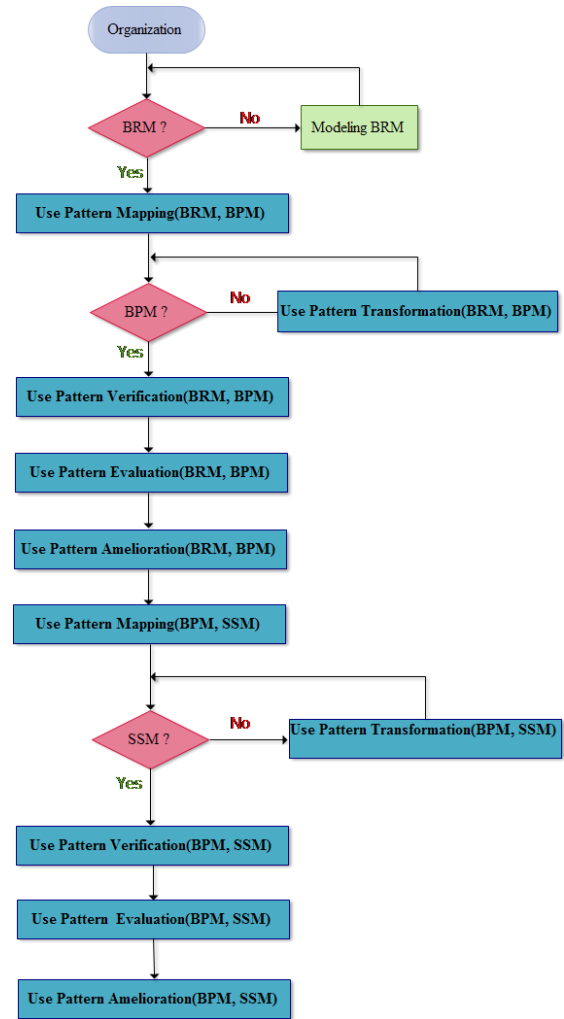
Table 6. Pattern of Amelioration

Fields	Values
Identification	Amelioration
Classification	Correction
Context	Required MPI, M1, M2, pattern of evaluation
Problem	We aim to propose instruction to correct the misalignment detected
Forces	Modifying the initial master plan, M1, and M2
Process Solution	<pre> graph TD AD((AD)) --> Instructions[Instructions to ameliorate MPI, M1 and M2] MPI[Consider MPI] --> Instructions M1[Consider M1] --> Instructions M2[Consider M2] --> Instructions Instructions --> MPR[Create MPR] Instructions --> M1R[Create M1R] Instructions --> M2R[Create M2R] </pre>
Requires	Pattern of Evaluation

Table 7. Pattern of Alignment

Fields	Values
Identification	Alignment
Classification	Aligning BR, BP, and SS
Context	Required BRM, pattern of mapping, pattern of transformation, pattern of verification, pattern of evaluation, and pattern of amelioration
Problem	We aim to align business requirement, business process, and software system
Forces	Allows all phases to align BRM, BPM, and SSM
Process Solution	In the case that BPM and SSM are not yet represented. 1) We use the pattern of mapping by considering business requirement and business process: Mapping(BRM, BPM) 2) We use the pattern of transformation between business requirement and business process: Transformation(BRM, BPM) 3) We use the pattern of verification between BR and BP. Verification(BRM, BPM) 4) We use the pattern of evaluation: Evaluation(BRM, BPM) 5) We use the pattern of amelioration: Amelioration(BRM, BPM) 6) We use the pattern of mapping by considering business process and software system: Mapping(BPM, SSM) 7) We use the pattern of transformation between business process and software system: Transformation(BPM, SSM) 8) We use the pattern of verification between BP and SS. Verification(BPM, SSM) 9) We use the pattern of evaluation: Evaluation(BRM, BPM) 10) We use the pattern of amelioration: Amelioration(BPM, SSM) When the BPM is represented we skip the level 2. When the SSM is represented we skip the level 7.
Uses	Pattern of Mapping Pattern of Transformation Pattern of Verification Pattern of Evaluation Pattern of Amelioration

Figure 2 presents a global representation of the guideline based on the pattern of alignment (Table 7).

**Figure 2. Global representation of the guideline**

5. CONCLUSION AND FUTURE WORK

This paper presents an issue related to the topic of alignment in organizations. The main question we try to answer is: How can organizations apply alignment? We deal particularly with the operational alignment and we focus on three levels: business requirement, business process, and software system. For this reason, we analyzed works that deal with these topics. As a result, we conclude that these approaches do not present a detailed guideline for helping organizations to apply alignment. This guideline is very important and may help organizations to exploit the different phases of alignment in a very easy way. Accordingly, we propose a pattern system to support this guideline. Our aim in the future is to work on a detailed description of the steps presented in this guideline. Moreover, we aim to apply our pattern system on different case studies to illustrate and validate our approach.

6. ACKNOWLEDGMENTS

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